

CHAPTER 1

INTRODUCTION

ROLE OF TECHNOLOGY IN DEVELOPMENT OF COUNTRY ECONOMY



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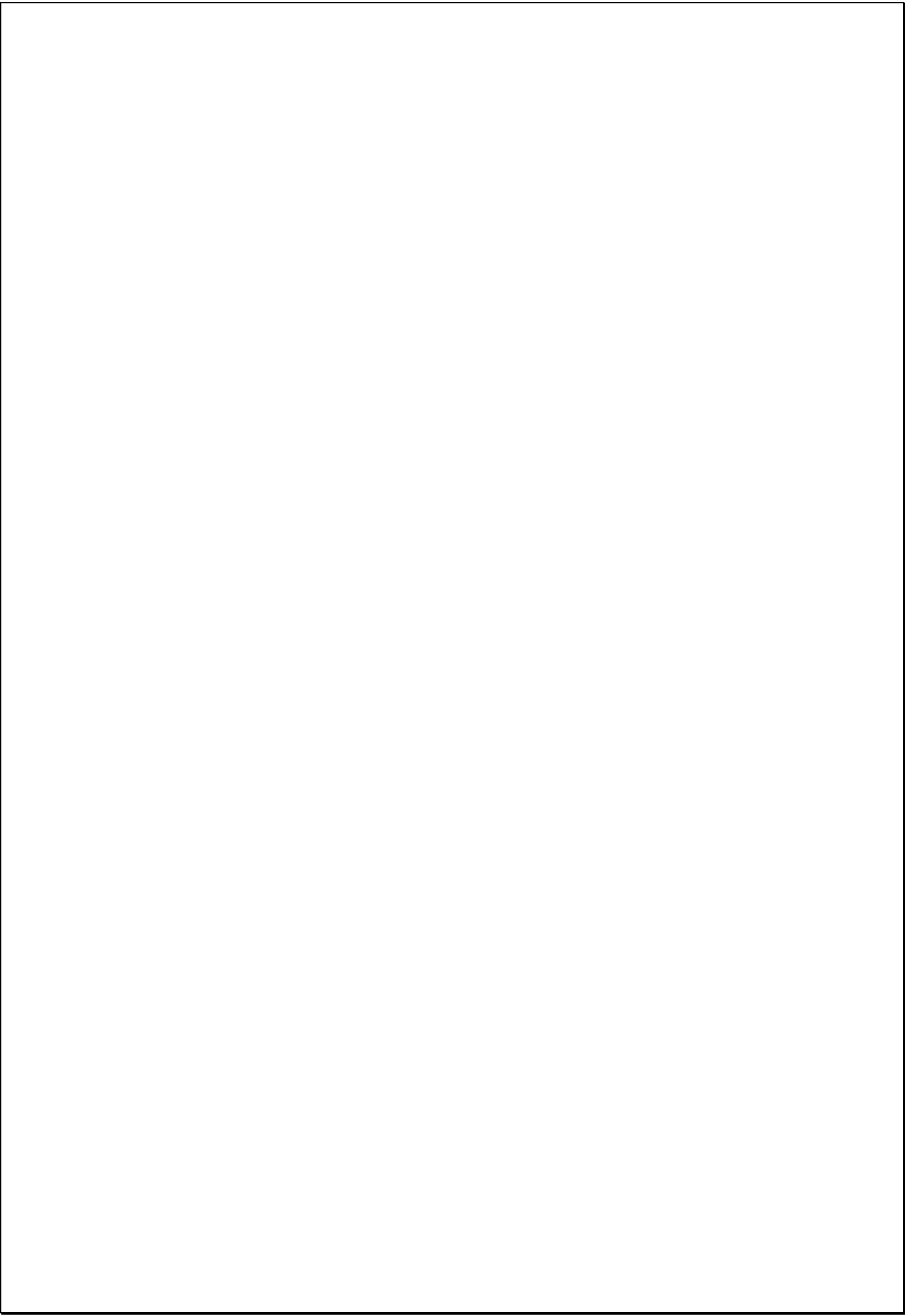
INTRODUCTION

Technology plays a crucial role in the development of a country's economy by fostering innovation, increasing productivity, and creating new industries. It drives economic growth through increased efficiency in production processes, improved communication and transportation systems, and the creation of new products and services. Additionally, technology has the potential to bring about social and economic transformation by providing access to information and education, promoting entrepreneurship, and empowering marginalized communities. Therefore, the incorporation of technology in a count. It can drive economic growth, create jobs, improve productivity and efficiency, and enhance living standards.

It helps to increase productivity, efficiency, and competitiveness in various sectors such as agriculture, manufacturing, and services. Technology also enables the creation of new industries and jobs, drives innovation, and provides access to new markets. Furthermore, technology can help address social and economic challenges, such as poverty, healthcare, and education. In order to fully reap the benefits of technology for economic development, countries must prioritize investment in r Technology plays a crucial role in the development of a country's economy by driving innovation and productivity. It helps in streamlining processes, reducing costs, increasing efficiency and opening up new markets. Technology also helps to create new industries and jobs, as well as improving access to education and healthcare. Additionally, technology can enhance governance and transparency, and contribute to overall economic growth and sustainable development. search and development, technology infrastructure

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ROLE OF TECHNOLOGY ON ECONOMIC

DEVELOPMENT

The new technology is codified knowledge in the form of routines and protocols. Technology helps to get enough knowledge about the use of economic resources to produce goods and services more efficiently. The economic growth has increased and is becoming efficient due to the advancement of technologies. In business, starting from production to the profit of the business got advanced due to technologies. It has also helped in spreading the business all over the world.

Technological advancement also brings a change in the total productivity of the business which is positive. A good change is good for the business for the productivity and the profit of the business. An improvement in technology results in a requirement for less costly inputs. Rapid growth can be achieved through high technology levels. Technology is the primary source of economic development, and the development of various technologies is very important for the growth of underdeveloped countries. Economic development also deals with the social well-being of people and the economic growth of market production.

Today's customer expects the same kind of experience in a professional environment that they have with technology in their personal lives. But for a business, this can be challenging to achieve. There have never been more choices, both in terms of how and where to deliver applications from, and who to partner with in the delivery of services.

Digital transformation provides a valuable opportunity for core business functions, such as finance and HR, to move away from manual processes and automate key areas like payroll, enabling leaders to focus on wider business opportunities.

Technology is having profound effects on labour markets. Automation and digital advances are shifting labour demand away from routine low- to middle-level skills to higher-level and more sophisticated analytical, technical, and managerial skills. On the supply side, however, equipping workers with skills that complement the new technologies has lagged, hindering the broader diffusion of innovation within economies. Education and training have been losing the race with technology.

Most major economies face the challenge of aging populations. Many of them are also seeing a levelling off of gains in labour force participation rates and basic education attainments of the population. These trends put an even greater focus on productivity and technological innovations that drive it to deliver economic growth.

HISTORY

Throughout history, technology has played a significant role in the development of a country's economy. For example:

The Agricultural Revolution: The development of new tools and techniques, such as the plow, led to increased agricultural production and a shift from hunting and gathering to settled farming communities.

The Industrial Revolution: The invention of machines, such as the steam engine, led to mass production and an increase in manufacturing and economic growth in countries like Britain.

growth of new industries, such as software development and e-commerce, and has transformed the way we do business and interact with each other.

In each of these cases, technology has driven economic development by enabling new forms of production, commerce, and communication, and has created new opportunities for countries. Throughout history, technology has played a major role in the development of countries' economies. For example, the industrial revolution in the 18th and 19th centuries brought about major technological advancements in transportation, manufacturing, and communication that led to significant economic growth and development in Europe and North America.

In the late 20th and early 21st centuries, the rise of information technology and the internet have transformed economies globally, leading to the growth of new industries and the digitization of many traditional ones. This has created new job opportunities, increased access to information and education, and facilitated the growth of e-commerce and the global marketplace.

In many developing countries, the adoption of technology has been a key driver of economic growth and modernization. By investing in technology and infrastructure, these countries can catch up to more developed economies and create new opportunities for their citizens. growth and, technological innovations have had a profound impact on economic growth and development. For example, the Agricultural Revolution of the 18th and 19th centuries, which introduced new farming techniques and tools, led to increased food production and allowed for population growth. The Industrial Revolution of the late 18th and early 19th centuries, which brought about new machinery and mass production methods, resulted in significant increases in manufacturing output and economic growth.

More recently, the Information Technology (IT) revolution of the late 20th and early 21st centuries have had a profound impact on the global economy, leading to the rise of the service sector and the growth of the digital economy. Technology has enabled businesses to operate more efficiently, reach new markets, and provide new products and services to consumers. It has also helped to create new jobs and industries and has improved access to information and communication

In summary, technology has been a key driver of economic growth and development throughout history and contain technology has played a key role in the development of country economies. In the 19th and early 20th centuries, the industrial revolution saw the development of new technologies such as the steam engine and the telegraph, which

transformed the way goods were manufactured and communication was carried out. These technological advancements led to increased productivity, economic growth, and the rise of new industries.

In the latter part of the 20th century, the digital revolution brought about by computers and the internet greatly impacted the global economy, leading to the development of new industries such as software and e-commerce, and increasing efficiency and interconnectivity in existing industries.

More recently, advancements in areas such as artificial intelligence, robotics, and biotechnology have the potential to bring about significant changes in the way economies operate, particularly in areas such as manufacturing, healthcare, and finance.

Overall, technology has consistently played a critical role in shaping the development of country economies throughout history and will likely continue to do so in the futures to play a vital role in shaping the future of global economy



FEATURE

INCREASED PRODUCTIVITY:

Technology can improve processes and increase efficiency, leading to higher productivity and output.

NEW INDUSTRIES AND JOBS:

The development of new technologies often creates new industries and job opportunities.

IMPROVED ACCESS TO EDUCATION AND HEALTHCARE: Technology can help to increase access to education and healthcare, leading to a more informed and healthier population.

ENHANCED GOVERNANCE AND TRANSPARENCY:

Technology can improve transparency and accountability in governance, leading to more effective and efficient governance.

GLOBAL INTERCONNECTEDNESS:

Technology has enabled greater interconnectedness between countries, allowing for easier trade and the sharing of ideas and resources.

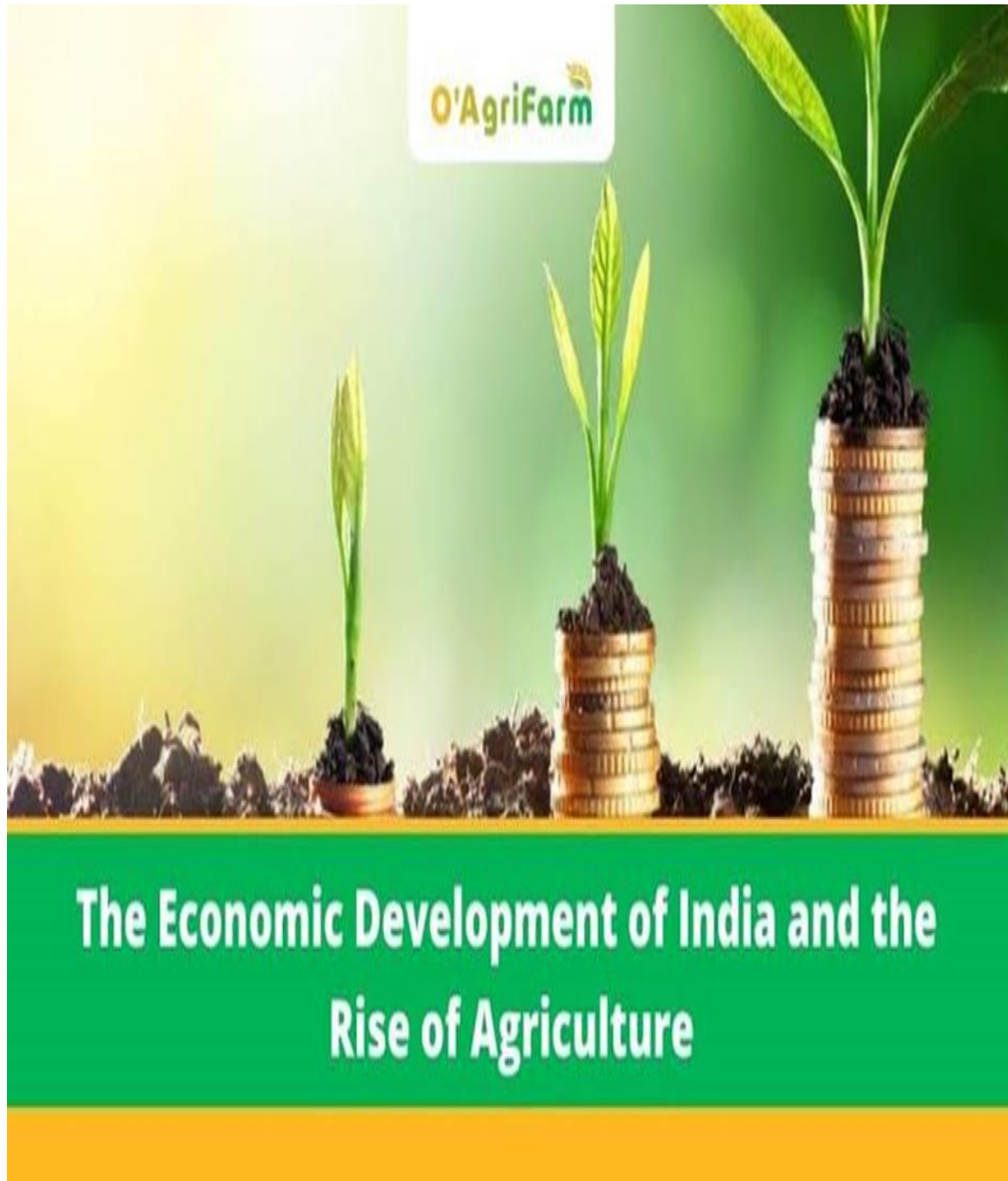
DRIVING INNOVATION:

Technology provides a platform for innovation and the development of new products and services, leading to economic growth and development.

COST SAVINGS:

Technology can reduce costs by automating processes, increasing efficiency, and reducing waste.

Overall, technology plays a critical role in driving economic development by improving productivity, creating new industries and jobs, improving access to education and healthcare, and enabling greater innovation and interconnectedness.



IMPORTANCE OF TECHNOLOGY IN ECONOMIC DEVELOPMENT

DISCOVERY OF NATURAL RESOURCES:

Modern technology helps human beings in utilizing the natural resources that are hidden in seas, lands, and mountains. If there was no advancement in modern technology then it was not possible to explore oil, iron, copper, gas, and gold.

AGRICULTURAL REVOLUTION:

The technology has helped to introduce fertilizers for plants and agricultural land, tractors, the invention of High Yield seeds, threshers, and pesticides. The revolution resulted in the cultivation of good crops and became a profitable profession and helped in putting an end to the shortage of food and grains.

EFFICIENT OPERATIONS:

Technology can optimize the operations of a company. Technology plays an important role in the generation of efficient processes. It can help you reduce or eliminate duplications, errors, and delays in the workflow, as well as accelerate the automation of specific tasks. Inventory technologies allow business owners to efficiently manage production, distribution and marketing processes. With the right technology in place, entrepreneurs can save time and money and make their businesses more productive and competitive.

GLOBALIZATION OF BUSINESS:

Technology in business made it possible for them to achieve a greater reach in the global market. Globalization has been carried out thanks to the wonders of technology. Anyone can now do business anywhere in the world. Technology has driven the development of electronic commerce, which has brought new dynamics to the globalization of companies. The diffusion of information technology has made production networks cheaper and easier and has been fundamental for economic globalization.

THE RISE IN PRODUCTIVITY:

Technology has helped in developing and producing at a higher rate in the economy. It has been achieved through proper utilization of human resources and using modern techniques in the production sector of the Indian economy.



IMPACT OF TECHNOLOGY ON ECONOMIC DEVELOPMENT

Economic development has always been influenced by technology. Cars, electricity and the internet are just a few technologies contributing to economic growth.

Technological advancements continue to grow at an extraordinary speed making it hard for anyone to deny their impact on the world.

Technology has been disrupting various industries and creating new markets. Therefore, the technological revolution is crucial in promoting economic development, especially in developing countries.

It improves efficiency and productivity, promotes competition and innovation, and provides opportunities for employment generation to millions of people worldwide.

Below are some ways technology has assisted economic development and will continue to help economic growth.

ECONOMIC GROWTH

The impact of technology on economic development is a critical component in understanding how technological change contributes to alleviating poverty and other social problems.

Although there is no one-to-one relationship between technological change and economic development, the association is essential.

Technology can affect growth by increasing productivity and expanding markets for goods and services. For example, improvements in agricultural technology have led to increased yields per acre and greater food security.

REDUCTION OF POVERTY RATES

Reducing poverty rates is a primary goal in developing countries, particularly those poor or emerging from conflict situations. New technologies have helped spread information about opportunities for jobs or education in other parts of the country or world. This helps people access the information they would not otherwise have had, increasing their chances of becoming part of the formal economy and helping them improve their lives.

AGRICULTURE DEVELOPMENT

Agriculture development is another area where technology can be beneficial because it allows farmers to grow more crops with fewer resources. For example, satellite imagery can help farmers plan their activities more effectively to be more efficient with their available resources; this leads to higher yields per acre for a given amount of land.

One area where technological innovation can be particularly beneficial in agriculture.

This has led researchers to develop better fertilizers and pesticides that help crops grow faster and produce more food for consumers.

Technology is an essential factor affecting economic development in Africa, which is why we should be a part of this development.

IMPACT OF TECHNOLOGY IN GLOBAL ECONOMY

The impact of innovations has been accelerating the trend towards urbanization all over the world that will help to make the country one step forward to being a developing country. The economic growth will increase due to the advancement of technology that will result in less poverty and portraying the country more clearly and will be able to compete with other developing countries. There is a need for development in technology in the 21st century otherwise it will not be possible to keep pace with other countries both technologically and economically and the effort of technological advancement is getting better with time.

The [impact of technology on global economy](#) also plays a very important role as it reduces the manpower in an organization and it also reduces the demand for low-skilled employees. Automation and digital advances also reduce the number of employees of an organization as they perform the tasks automatically. The wage inequality was also identified due to the implementation of new technologies. While monetary inequality has increased in many countries, inequality between countries has been declining as the fast-growing economy is still narrowing the money gap through the high-income economy. Technology poses new challenges to this economic integration. Productivity-driven growth in the emerging economy has been a major factor in the partnership, thanks to their comparative advantage in hard-earned production from their large pools of low-skilled, low-wage workers. Such comparative advantages are eroded by the low self-employment of low-skilled workers, creating the need to create alternatives to growth in line with technological change. Transformation is also needed at the global level, although the key component of the technology – and global trade – the system is more efficient and all in line with national policy. Not only should the past benefit from establishing a system of international law-protected trade protected from the wind, but new sectors should be established in the next phase of global trade led by digital flow to ensure open access and fair competition. Reasonable immigration policies can complement national policies, such as pension reform and lifelong learning, in reducing the effects of aging. The growth in national income in developed countries cannot be attributed solely to finance. Kindle Berger noted that much of this growing production is due to technological change. Robert Solow

estimated that technological change slowed about 2/3 of U.S. economic growth. after allowing for staff growth and financial stock. Get to know much detail on global trade from [term paper help](#), a brilliant team of Source essay anytime.

Technology can be considered a major source of economic development and various technological changes contribute significantly to the development of developed countries. The second figure shows that innovation is saving workers and R's shows that the same product can be produced with less input but labour savings are greater than in the capital. The third figure shows that this innovation saves money and R's shows that the same output can be produced with less input after a technological change but saves more money than workers.

It is thought that technological advancement is far more important than capitalization. But monetary policy alone can bring economic development to a limited extent and progress can be halted if there is no technological change. The country cannot always rely on technology imports. The nation that spends the most on scientific and technological research will tend to grow faster than any other country that collects more money but spends less on technology.



HOW TECHNOLOGY IS PLAYING ROLE IN REVIVING INDIA ECONOMY

According to a recent report by the World Bank, the Indian economy is set to expand by 8.3 per cent in 2021, 7.5 percent in 2022 and by 2023, India is expected to grow at 6.5 percent. The economic impact of COVID-19 pandemic in India has been quite disruptive and there has been a setback. The digital era has brought unprecedented change to technology, business, society and the situation have started to break the inertia of digital adoption and cloud will continue to be critical as part of the evolution. Business across verticals have been affected from the global crisis, but there are certain industries which are showing the sign to change the game for the economy in the years to come.

Industries like healthcare and pharmaceuticals, BFSI, e-commerce and retail and manufacturing are early adopters of emerging technologies. Organizations in these industries have reacted to difficulties introduced by the pandemic and are focusing heavily on creating strategies in the new normal to up their game in next five years. The string that ties together in all the four industries is modern technology and cloud plays a role to help all these four industries catapult into a game changer for the Indian economy. New-age technologies will be giving India a chance to carve itself a unique identity as a global hub for cloud solutions.

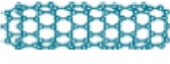
Industry 4.0, Digital supply chain, digital twins, Digital Assistance etc. are some of the technologies that will help in the economic revival by 2025. Emerging technologies are beginning to enable industries to rebuild India's economic status in a post Covid world and these industries will be the real game changers in the time to come. Companies related to these verticals are embarking on their journeys and starting from different points to use smarter products, smarter manufacturing and smarter supply chains. The smarter organizations are leveraging the huge potential of emerging technologies to their advantage, and once the economy bounces back, these companies will be one of the frontrunners and will capture a bigger percentage of the market.

Agrees Prasad Rai, Vice President, Global Strategic Clients Group, Oracle India, "With the Indian economic contraction since the last year, the economy is looking forward to revive and many industries in India have shown the potential to be game changers in the nation's economic development. The role of these game-changing corporations is going to be pivotal for the recovery of the Indian economy post-pandemic. We have seen that through the pandemic, data and analytics became increasingly essential. Industries that had well defined processes based on year of historical data found themselves at a complete loss since that data was no longer relevant when everything changed overnight. They needed to quickly analyse new data and employ artificial-intelligence and machine-learning to "re-learn" what was the new normal in customer behaviour. On the contrary, organizations that have invested in digital resources quickly were much more secure and prepared even as the country was hit by a huge second wave."

Prasad Rai observes that the potential game-changers from sectors like healthcare and pharmaceutical, banking, manufacturing, and retail and e-commerce have progressive technologies like modern cloud, AI, ML, IoT, Blockchain, and big data analytics at the heart of their transformation.

Exhibit E1

Twelve potentially economically disruptive technologies

	Mobile Internet	Increasingly inexpensive and capable mobile computing devices and Internet connectivity
	Automation of knowledge work	Intelligent software systems that can perform knowledge work tasks involving unstructured commands and subtle judgments
	The Internet of Things	Networks of low-cost sensors and actuators for data collection, monitoring, decision making, and process optimization
	Cloud technology	Use of computer hardware and software resources delivered over a network or the Internet, often as a service
	Advanced robotics	Increasingly capable robots with enhanced senses, dexterity, and intelligence used to automate tasks or augment humans
	Autonomous and near-autonomous vehicles	Vehicles that can navigate and operate with reduced or no human intervention
	Next-generation genomics	Fast, low-cost gene sequencing, advanced big data analytics, and synthetic biology ("writing" DNA)
	Energy storage	Devices or systems that store energy for later use, including batteries
	3D printing	Additive manufacturing techniques to create objects by printing layers of material based on digital models
	Advanced materials	Materials designed to have superior characteristics (e.g., strength, weight, conductivity) or functionality
	Advanced oil and gas exploration and recovery	Exploration and recovery techniques that make extraction of unconventional oil and gas economical
	Renewable energy	Generation of electricity from renewable sources with reduced harmful climate impact

SOURCE: McKinsey Global Institute analysis

TYPES OF TECHNOLOGY

1. ARTIFICIAL INTELLIGENCE

Artificial intelligence is the simulation of human intelligence by computer systems and other machines aimed at performing higher-level tasks that require human intelligence and are normally performed by human beings.

Smart assistants on our devices, such as Siri, Google Assistant, and Alexa are some of the most popular examples of AI. Scientists make continuous efforts to improve AI technology and make it as closely intuitive to human intelligence as possible.

2. INFORMATION TECHNOLOGY

Information Technology, also known as IT, involves the use of computer systems—hardware and software— and telecommunications for the purpose of storing, sending, and retrieving information. This technology finds its applications in resolving business challenges and helps streamline organizational processes.

Some significant examples of IT are performance management software, content management software, and telephone and radio equipment.

3. SPACE TECHNOLOGY

Space technology is a field that helps space scientists plan and execute space exploration, spaceflight, and the observation of Earth. It is mainly used for communication and navigation and to support human activities beyond Earth's atmosphere.

Telescopes, GPS systems, radio and imaging, space traffic management, and low-Earth orbit satellites are some of the most notable contributions of space technology toward making space exploration accomplishable.

4. ENTERTAINMENT TECHNOLOGY

Entertainment technology aims at providing better entertainment experiences to users using science and technology. It caters to different subfields of the entertainment industry, such as television, gaming, advertising, streaming, animation, etc.

Some of the common examples of entertainment technology used for various purposes include virtual reality, augmented reality, interactive environments, automation, simulations, and audio-visual areas.

5. MEDICAL TECHNOLOGY

Medical technology aids in more efficient and accurate health assessments, diagnosis, management of health ailments, and treatment, along with implications in the field of pharmacology.

Medical technology has a prominent place in the healthcare industry through techniques, such as robotic surgery, therapeutic devices, biometrics, and diagnostic imaging.

6. OPERATIONAL TECHNOLOGY

Operational technology, popularly known as OT, is the use of [software and hardware](#) for controlling industrial equipment in order to carry out security measures. This technology enables monitoring, controlling, and detecting changes in the processes, devices, and events of industrial networks.

Fire control systems, physical access control mechanisms, and building management systems are some instances of operational technology useful in industrial and building equipment.

7. ASSISTIVE TECHNOLOGY

Assistive technology makes various daily activities, such as bathing, walking, hearing, movement, and climbing stairs convenient. This technology has the widest application for individuals with disabilities, thus aiming to improve their quality of life.

Some of the most common and widely used assistive technologies include artificial limbs and arms, wheelchairs, walkers, and crutches.

8. COMMUNICATION TECHNOLOGY

A crossover of information and communication, communication technology deals with the transfer of information in the form of messages among machines and people. It contributes to the decision-making process and solving problems in people.

Communication technology has made huge strides in recent times, with some examples of this technology being virtual assistants, social media platforms, Wi-Fi, and Bluetooth.

9. HOSPITALITY TECHNOLOGY

This technology aids in client interactions in the hospitality sector using technological infrastructures, such as [software](#) and networking means. It has various applications, such as making reservations, offering assistance in various languages using machines, and other areas where it facilitates off-site assistance to add convenience to client interactions.

10. SUPERINTELLIGENCE

The use of artificial intelligence and computer systems to facilitate the betterment of human life is referred to as superintelligence. This form of technology replaces various operations that humans would have to do otherwise accomplish manually. Some examples of superintelligence are chatbots, virtual agents, and [self-driving cars](#), among others.

11. BUSINESS TECHNOLOGY

The application of technology and science to provide high-level assistance in business activities is known as business technology. This technology has a very wide scope percolating down through different areas of business management, such as information technology, digital marketing, data management, and E-commerce technology to name a few.

12. AGRICULTURE TECHNOLOGY

[Agriculture technology](#) is helping replace many manual operations in agriculture that can often be complex and time-consuming with the help of science. Farmers who handle large-scale agricultural production and management are often equipped with modern agricultural technology, such as machines that help regulate temperature and moisture and GPS technologies.

13. ROBOTICS TECHNOLOGY

A combination of science and mathematics, [robotics technology](#) is associated with developing machines and software to manufacture intelligent robots. Such robots are used in various other industries. One of the industries that widely use this technology is healthcare, which has immensely beneficial medical robotics implications.

14. EDUCATIONAL TECHNOLOGY

EdTech, or educational technology, has revolutionized the teaching and learning processes with breakthroughs, such as computer-based teaching, interactive learning tools, audio-visual systems, and online learning resources. Educational technology mainly aims to improve the learning process by replacing or complimenting the more traditional approaches with scientific and technological tools.

15. PRODUCT AND FOOD TECHNOLOGY

Product and food technology aims to improve and automate various manufacturing operations in the food industry, which is very beneficial to food businesses involved in mass production. Whether it is the automation of manufacturing a food product or quality testing and control, there are various types of machinery involved in the food production process— all thanks to food product technology.

16. ARCHITECTURE TECHNOLOGY

Today, architecture is so much more than the mere process of designing structures. It involves eco-friendly buildings, space-saving approaches, green buildings, and much more. Architectural technology, while helping architects with the more conventional designing processes, also extends to creating better ways for architects to understand different elements of the design and the geographical considerations while creating sustainable and practical models.

Some examples of architecture technology are Building Information Modelling (BIM), renderings, generative design, 3D printing, and IoT.

17. CONSTRUCTION TECHNOLOGY

The construction industry benefits greatly from construction technology, which is the application of computer systems and software to the construction processes. While it has a certain level of overlap with architecture technology, some common types of technological aids used in construction are drones, virtual reality, data collection apps, and various forms of AI to automate certain processes.

18. BLOCKCHAIN TECHNOLOGY

[Blockchain technology](#) is a web-based monetary system that allows information to be encrypted. While it was initially created for digital assets, this technology is currently used to create applications for online stock exchanges and even social media sites.

19. AEROSPACE TECHNOLOGY

Aerospace technology is one of the most important technological advancements, which finds its applications in the field of space flight, aerospace engineering, and airplane designing. With the advent of this technology, we have access to innovative ways to use space technology and develop components that are more efficient, such as wings for spacecraft and airplanes with a high form of precision and accuracy.

20. ENVIRONMENTAL TECHNOLOGY

With the need of the hour being the conservation of resources and our ecosystem, environmental technology has received significant focus, which involves the use of scientific knowledge and techniques that help improve the environment. Some common examples of environmental technology are wind turbines, solar panels, wastewater purifiers, and photovoltaics.

21. BIOTECHNOLOGY

Biotechnology deals with the use of living organisms to create different types of products. It is based on biology and makes use of biochemical and cellular processes to come up with technologies. Biotechnology aims to improve our lives. Some of the modern-day biotechnological processes are genetic engineering, stem cells, and tissue engineering, transgenesis, and DNA cloning, among others.

22. VEHICLE TECHNOLOGY

Vehicle technology involves vehicle development, wherein the internal parts of cars and aircraft are upgraded from time to time for improved efficiency. It also aims at bringing down carbon emissions while enhancing the comfort of the vehicle. Some of the notable vehicle technologies are parking sensors, collision avoidance systems, and the most recent development— self-driving cars.

23. FORENSIC SCIENCE TECHNOLOGY

An important area to add more precision to criminal justice, forensic science technology helps identify facts in the investigation process with the help of scientific knowledge surrounding the elements of evidence. Fingerprint analysis, DNA profiling, and facial reconstruction are some of the most remarkable strides in forensic science technology.

24. MILITARY TECHNOLOGY

Military technology aims at creating better tools, weapons, and equipment to better equip military personnel for wars. With heavy investment and the need for countries to outpower each other, some of the notable technological advancements in this area include nuclear weapons, radar equipment, and advanced guns and missiles.

25. SPORTS TECHNOLOGY

Sports technology is the use of scientific knowledge to design and implement technologies for sports. This technology has helped take sports to the next level with innovative training techniques and designing better competitive surroundings. Video assistant referees, flying drones, camera movements, and edge detectors are among the most popular and commonly used technologies in sports.

26. ELECTRICAL ENGINEERING TECHNOLOGY

EET, or electrical engineering technology, combines science and the principles of electrical engineering for improved design, installation, operation, and maintenance of electrical systems. Some of the products of EET are transmission lines, lighting systems, power plant generators, microsensors, and Nano sensors.

27. 3D PRINTING TECHNOLOGY

3D printing technology makes the task of printing objects in three dimensions achievable and more efficient. One of the most prominent technologies for this purpose is CAD software, which helps design the object, which is then entered into a printer to create the object.

28. QUANTUM TECHNOLOGY

A subfield of physics, quantum technology has various applications in fields, such as communication technology, electric and magnetic sensors, quantum computing, modelling, metrology, and cryptology.

29. INDUSTRIAL AND MANUFACTURING TECHNOLOGY

This form of technology helps improve industrial and manufacturing processes and operations with scientific techniques. Some of the common technological advancements in this area include automated machines, large and efficient furnaces, conveyor belt systems, and CNC milling machines.

30. MARINE TECHNOLOGY

Marine technology helps come up with sophisticated equipment for safe use, protection of, and intervention in the marine environment. Digital route management of ships, integrated control systems, smart defence technology, and IoT on vessels are some of the most widely used modern marine technologies.

ADVANTAGES

INCREASED PRODUCTIVITY:

By automating various processes, technology can increase efficiency and productivity, leading to increased output and economic growth.

JOB CREATION:

The adoption of technology can create new job opportunities in areas such as software development, data analysis, and e-commerce.

IMPROVED COMPETITIVENESS:

Advanced technology can help companies to compete more effectively in the global market, boosting the country's competitiveness.

BETTER ACCESS TO INFORMATION:

Technology can provide better access to information and knowledge, enabling individuals and businesses to make informed decisions, leading to increased innovation and economic growth.

IMPROVED STANDARD OF LIVING:

Technological advancements can lead to the development of new products and services that can improve the standard of living for citizens.

In summary, technology can bring significant benefits to a country's economy by driving productivity, job creation, competitiveness, access to information, and improved standard of living.

DISADVANTAGES

JOB LOSS:

Automation and artificial intelligence can replace human workers, leading to job loss and unemployment.

INCOME INEQUALITY:

The increasing use of technology can result in widening income gaps between highly skilled workers and those with low or no technical skills.

DEPENDENCE ON TECHNOLOGY:

Countries may become too dependent on technology, making their economies vulnerable to disruption in the event of technological failures or cyber attacks.

DIGITAL DIVIDE:

Access to technology is not equally distributed, which can result in a digital divide between those who have access and those who do not, perpetuating socio-economic inequalities.

ENVIRONMENTAL IMPACT:

The production, use, and disposal of technology can have negative environmental consequences, such as e-waste and carbon emissions.

REMEDIES FOR ECONOMIC DEVELOPMENT-

- **Growth Of Export-**

The production of goods that have a comparative advantage in their production will solve the limited market size.

- **Industrialization-**

The improved equipment of production helps in raising agricultural production and the establishment of the industries helps in meeting the basic needs of all business sectors in the economy.

- **Self-Management Strategy-**

The country system needs to be adopted to coordinate and to help with the production of economic development in the country.

Impact on technological innovations in Sustainable economic growth-

- Education.
- Agriculture.
- Health.
- Construction.
- ICT.

ACHIEVEMENT THROUGH TECHNOLOGIES-

- The aim of maximizing the outputs.
- Reducing the cost of production
- The aim is to attain higher growth rates.

The role of technology in economic development got more importance due to the labour shortages in the Western countries which were resulting in a lack of production.

The development and growth are different but both of them are needed when it focuses on economic development. It is helping in the production of goods all over the country. Even the technology has introduced online applications through which there are sizes and specifications according to the choices of everybody all over the world. We can sit at home and buy things that we want that will increase the profit and also will bring growth in the economy of the country. Social health also develops by increasing the wealth and the income levels in the economy which can also create problems on the other side. Schumpeter observed that technology innovation is the only determinant of progress in the economy. He was motivated by the fact that economic growth and the innovation of technology are closely related. The theory says that if one economy is the technology innovator and then another is a technology adopter and will maintain lead in per capita related to another one.

REFRANCE

www.economicdiscussion.com

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www.wikipedia

www.quora.com

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Sehgal foundation

Express computers

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CHAPTER 2

REVIVE

OF

LITERATURE

2.1 MEANING OF LITREATURE

The literature review is a written overview of major writings and other sources on a selected topic.

A literature review is a piece of academic writing demonstrating knowledge and understanding of the academic literature on a specific topic placed in context. A literature review also includes a critical evaluation of the material; this is why it is called a literature review rather than a literature report.

A “**review of the literature**” is a classification and evaluation of what accredited scholars and researchers have written on a topic. Occasionally you will be asked to write one as separate assignment, but often it is part of the introduction to an essay, research report, or thesis. In writing the literature review, your purpose is to convey to your reader what knowledge and ideas have been established on a topic, and what their strengths and weaknesses are. As a piece of writing, the literature review must be defined by a guiding concept (e.g., your research objective, the problem or issue you are exploring, or your thesis). It is not just a descriptive list of the material available, or a set of summaries. A literature review consists of an overview, a summary, and an evaluation (“critique”) of the current state of knowledge about a specific area of research. It may also include a discussion of methodological issues and suggestions for future research. Besides enlarging your knowledge about the topic,

2.2 NEED OF LITERATURE REVIVE IN RESEARCH

There are several reasons why a review of literature is important:

To gain an understanding of a topic: A review of literature can provide an overview of the current state of research on a particular topic, helping researchers gain a deeper understanding of the subject.

To identify gaps in knowledge: By reviewing the existing literature, researchers can identify areas where further research is needed and design their own studies to address these gaps.

To inform study design: A review of literature can help researchers to identify the strengths and weaknesses of previous studies, enabling them to design their own studies more effectively.

To establish the foundation for a research project: A comprehensive review of literature lays the foundation for a research project by providing context, background information, and a basis for comparison for the new research.

To provide context for new findings: A review of literature can help to contextualize new findings and place them in the larger body of research on a particular topic.

To establish credibility: A thorough review of literature demonstrates that the researcher has conducted a comprehensive search of the relevant literature and has a solid understanding of the existing research on a particular topic.

To identify potential sources of funding: By reviewing the literature, researchers can identify areas where funding may be available for their research projects.

Overall, the review of literature is an important step in the research process that helps researchers to better understand the subject they are studying, design their studies more effectively, and ultimately, contribute to the advancement of knowledge.

2.3. REVIVE OF VERIOUS LITRETURE

2.3.1 MR. ERIK J .DE BRUIJN and HARM-JAN STEENHUIS in their research paper present the status of “TECHONOLOGICAL COMPETITIVENESS OF ECONOMIC DEVOLAPOMENT “

Technology is considered as one of the key factors that influences economic development. As such, many authors in the Field of technology management mention this impact on economic development as a motivator for their technology-oriented study. But what is this relationship between technology and economic development? How? does technology lead to economic development? Another way of looking at technology and development is by looking at how technologically. competitive countries are. This would give insight into what different. countries do that makes them competitive. In this Feld there are at least two. interesting publications that compare the competitiveness of nations: the IMDs World Competitiveness Yearbook and the World Economic Forum's Global Competitiveness Report. For example, the Global Competitiveness Report (GCR) has a focus on economic performance and looks at factors that contribute to this performance. [Schwab (2009)]. In particular, the GCR distinguishes 12 pillars of competitiveness: institutions, infrastructure, macroeconomic stability, health and primary education, higher education and training, goods market efficiency, labour market efficiency, financial market sophistication, technological readiness, market size, business sophistication, and innovation. Technology has an influence in several of these areas such as through technological readiness and innovation but also through institutions (property protection which relates to technology discussion and commercialization), infrastructure (for example number of telephone lines, which can be viewed as availability of a specific technology), higher education and training.

(Internet access in schools), and business sophistication (production process sophistication). The GCR further established that different pillars have a different influence on economic performance based on the stage the country is in. The GCR distinguishes three different stages, that is, factor driven economy, efficiency driven. economy, and innovation driven economy. The emphasis in the factor driven stage is. on institutions, infrastructure, macroeconomic stability, and health and primary education while in the innovation driven stage the emphasis is on business sophistication. and innovation [Schwab (2009)]. Thus, it can be deduced that the type of technology which is beneficial for countries depends on its stage.

As per my revive it has allowed a better understanding of technology different component That together providing the components works in development of country economy at firm level to the macro i.e., nation level .it contributes in gross domestic product of nation. when it applied at national level it leads to the national technological and economic development but .it is not clear whether these measurements related to real-world or theoretically abstraction

2.3.2 I A MAGOMEDOV H A MURZAVE and A M BAGOV IN their research present “HOW DIGITAL TECHONOLOGY AFFECT ECONOMY ”

This article details how digital and information technologies affect the economy of modern society. Today, with no doubts we entered an era of rapid change and, most likely, soon, the most significant areas of human activity (economic, managerial, scientific, security) will take on completely different formats and meaning. People are changing, and their views on social life too. Digital technologies in the economy can be interpreted as information technologies that make possible to optimize the work of e-business structures in the current economic situation. The digital economy can be classified into three main elements: elements of infrastructure (equipment and programs, telecommunications devices and more), the direction of electronic business, the direction of electronic commerce (trade in goods online)

A feature of structural dependence is the impossibility of overcoming it, since progress in the field of digital technologies is proceeding at a fairly high speed, and new technologies can be reproduced only on the basis of previous results. If the country does not possess them some technical and technological solutions are lost, then it is impossible to create something new and take the next step. For this reason, the state of the digital sector, in particular, its elemental base, special technological equipment that provides the necessary parameters for microcircuits, is a determining factor in the development prospects of the whole society.

As per my revive digital technology are consistently expanding their areas to use in economy the digitalization are increasing consistently similarly the development and subsequent implementation of digitalization plan are accepted by every large firm the most important is technology is considered as digital platform. the digital technology is innovation of industries in today economy the digital sector companies is increasing and that becomes the strength of the economy for growth of country.

2.3.3 Ahmed Muayad Younus in their articles present “technological advancement and economic growth ”

From the beginning of human history, technology has progressed and has unceasingly continued to evolve. During the twenty-first century, technology has evolved into a complex structure that count Economic growth is described as an increase in the number of tools and products that will be utilized to address the needs of people in any country or region, regardless of their location. An economic growth rate can be determined by asking whether there has been a real increase (excluding price increases) in (Gross Domestic Product) from one year to the next. is defined as the market equivalent of all measurable values produced by a single economy and is used to calculate the rate of economic growth. The accumulation of capital, technological advancements, and an expansion in the population and labour force are the three most important indices of economic growth. aims vast amounts of data. The source of economic growth, on the

other hand, has shifted over history. The expansion and strengthening of the state's coffers were seen to be the foundations of mercantilism, and the requirement that export volume exceeds import value was regarded as a general norm for economic development. The physiocracy was implemented. The concept of maintaining the natural order is a popular one. Agriculture, as the foundation of the economy, is the only means of achieving long-term growth. This way of thinking acknowledges the necessity of industry but does not place any emphasis on the role of commerce. The classical growth theory was established by A. Smith, R. Malthus, and D. Ricardo between the end of the 18th century and the beginning of the 19th century. This theory is based on the idea that the degree of income per person has a direct relationship with population growth.

As per my revive technology differences are explained inter activity in equality of the income of country. technology makes business and induvial to use technology more efficiently. technology advancements varying degree of both micro level and macro level, to achieve this goal, the volume of commodities and services generated in the society must be increased. technology is not only a means of increasing the riches and prosperity of nations, but it also provides the people with the power to do things that they have not been able to achieve in the past because of technological advancements.

2.3.4 HULYA KESICI CALISKAN in their articles present “ technology change and economic growth ”

Technological transformation does not only mean a way to increase the richness and wealth of the nations, but it also gives the people the ability to do things that they have not done so far. Therefore, the novelty determines whether a whole life quality will develop positively or negatively (Freeman and Soete, 2003). As a matter of fact, negative consequences of technological transformation and industrialization – for example rapid consumption of natural resources, pollution of natural environment, disruption of theological balances- are greatly restricting the chances and sources for the future generations. Rapid consumption of the resources poses the threat for big hostilities and conflicts among the nations in the future. Even at this point, it is obvious that the cost to replace the rapidly consumed resources will be very high and economic problems will be inherited to the future generations.

If we examine the countries of the world, we can see those economic differences among the countries cause important differences in life quality. While certain countries regularly improve the life quality of individuals with the support of advanced technologies, other countries is way behind them in this improvement. It seems that economy has been guiding the technology as the innovations introduced to the world by technological advances are closely correlated with economy and follow the economic relationships. The nations that could efficiently disseminate technology and information to all areas of the society can create new areas of employment in their countries. However, these new areas require qualified work force. Thus, necessary revisions should be made to the education policies to ensure the development of human sources with such qualifications supporting the economic growth.

As per my revive the new technology brings economic developments technological development eliminated certain jobs and work areas and made a negative impact to employment As a result of e-commerce made on Internet, the dimensions of commerce have

changed. The producers and consumers could meet with each other in international markets through e-commerce and make commerce. Increasing mutual dependency between the global countries make economies exposed to the financial crises. One of the areas in which we can observe the effects of technological development is the financial markets.

2.3.5 Dhanshri M. Patil in their articles present “ RECENT TRENDS OF PRINT MEDIA IN DEVELOPMENT COMMUNICATION

communication technologies which would improve the business Media like print media and electronic media plays a significant role in development communication of country. Even after the advent of electronic media-like radio and television, the print media has not lost its charm or relevance. Print media has the advantage of making a longer impact on the minds of the reader with more in-depth reporting and analysis. The contribution of print media in providing information and transfer of knowledge is remarkable. Now-a-days, print media is faster than all ever before due to amazing advances in 3 technologies in recent years. The scientific and technological advancements have brought about steady fast development in the media world. New media are coming up while the old ones are being improved upon and in this process their availability has increased manifolds. They are now conquering even the remote and distant regions of the world.

The eighties and nineties saw the growth of medium Indian language newspapers. They adapted the latest printing and communication technology to bring out multiple editions. today the situation has going to change drastically, and new advent technology are used in to the all over world. The best result of such technology is electronic newspapers (on-line versions of city newspapers), which started in the late 1990s, have expanded to the point that now thousands of the world's newspapers are now on the Internet. It has a major benefit that publishing on-line newspaper saves the publishers two of their biggest expenses: newsprint and distribution costs. But on-line articles are not as in-depth as the print versions. And hence the print media plays a significant role even if there is a stiff competition from various electronic media. The ad revenue trend: The press is still the dominant medium for advertising in the country, even if television has steadily increased its share. The global story of ad revenue growth for paid-for dailies in 2006 and over five years was somewhat more cheerful.

As per my reive Today, due to technology faster evolution in India the circulation trend of newspaper showing an increasing trend as compare to the pre- independence era. Increasing circulation trend of newspaper in India shows that it wins to stay in the stiff competition faced from the electronic media. the print media has its lion share in the development communication of both developed and underdeveloped countries.

2.3.6 Ivo De Loo and Luc Soete in their research paper present “ The Impact of Technology on Economic Growth: Some New Ideas and Empirical Considerations”

The relationship between technology and economic growth has been captured in a vast number of formal models for almost half a century. Early neoclassical models like Solow (1956) treated technical change as an exogenous variable. R&D based models relating technical change and economic growth have been unsuccessful in explaining the recent productivity paradox. Jones (1995a) gives several explanations for the contrasting relationship between the state of technology and productivity growth, known in the literature as the productivity paradox while firms postpone their most risky R&D efforts and invest in variations on existing products in order to fulfil consumers' demands. Besides, they might consider financing their R&D efforts more and more by themselves, instead of speculating on financial markets. If this way of reasoning is in line with economic practice, then there should be a relationship between the amount of money spent on R&D and the real interest rate. Equivalent innovations as well as non-equivalent innovations utilize R&D, but the latter type of innovation is probably patented more heavily by default. This means that analysing the ratio of patents granted by the USPTO and total R&D expenditures (in constant prices) can test the notion. The ratio is standardized so that it equals 1 in 1985. For reasons of data availability, we focus on total manufacturing (ISIC code 30) only. Figure 3.1 contains the relevant statistics. The countries considered are the G7, Japan, the Netherlands, the US, Germany, France and the UK.

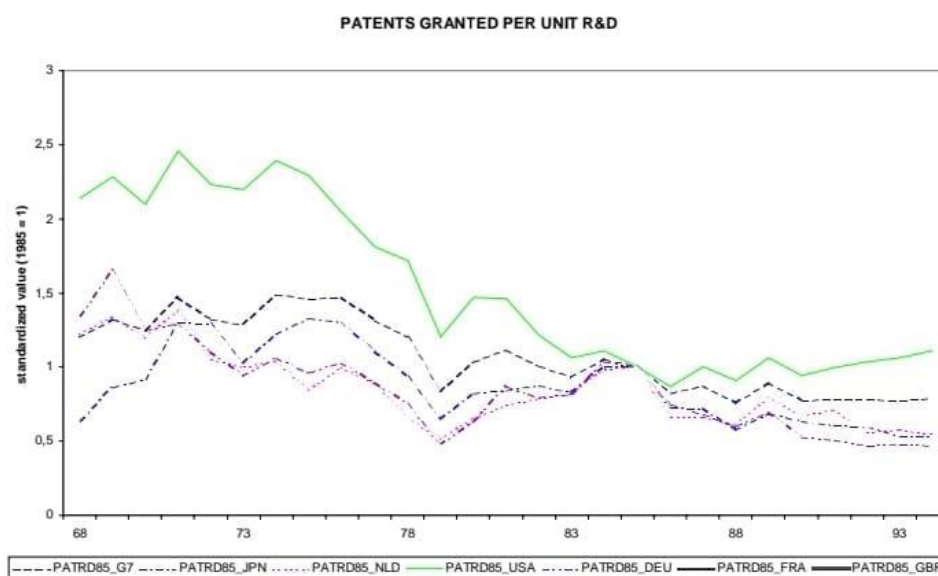


Figure 3.1 Standardized value of the ratio between patents granted by the USPTO and deflated R&D expenditures for several countries between 1968-1994 (1985 = 1). *Source: USPTO, ANBERD and UNIDO databases.*

As per my revive R&D based models relating technical change and economic growth have been unsuccessful in explaining the recent productivity paradox: R&D efforts have risen continuously in advanced countries during the post-war period whereas productivity growth has, if anything, declined. The notion that R&D efforts are more and more attributed to product differentiation, thus enlarging consumers' welfare while simultaneously exhibiting only limited effects on economic growth, may very well be the most promising in explaining the productivity paradox.

2.3.7 Debdas Banerjee in their research paper present “Science, Technology and Economic Development in India: Analysis of Divergence in Historical Perspective”

the increase in the national power to create science has failed to turn out as the remedy for technological backwardness. The weakness becomes revealing not so much when technology is developed in a non-competitive "space" (e.g. public sector enterprise) as it is when it comes to the development of commercially viable as well as competitive technology. In other words, the transmutation of scientific knowledge into field-level practices' is often found to be difficult.

In the neoclassical trade theory. "Technology is treated to be freely available to all countries and, within countries, to all firms. Countries simply settle on appropriate levels of capital/labour intensity in accordance with their factor price ratios. determined by their relative endowments of physical capital and labour. Even the appropriate technology' literature assumes that efficiency would reign in all firms if only the relevant techniques were "discovered and factor price distortions removed Nelson 1981]. This sort of assumption induces the transfer of technology literature to focus attention on the terms of the transfer and imperfections in international technology markets, rather than on the efficiency with which the technology is used by the importer. The later endogenous neoclassical models, usually referred to as the "New Growth Theory' (NGT) leg. Lucas 1988; Rebelo 1991; Romer 1990] emphasises the accumulation of human capital (by means of human capital) as determinant of the rate of economic growth. The difference between the developed and the developing countries is one of difference in the respective sizes of human capital. If this be the case, one would wonder why India despite having a considerable size of human capital (scientists, engineers and technicians) that compares well with many of the developed countries.

As per my review the weakness of harnessing science to commercially viable and competitive technology is better explained by the 'receiving system' than the human capital formation as postulated in the 'New Growth Theory'. The technological rule provides clues to the processes of disjunctions between pure and applied science and, between research and development. The weakness of the post-colonial state to politically resolve crucial economic issues and undertake strategic development policies gave way to the 'technocratic' solutions that entrench the earlier trend

2.3.8 S. Chandrashekar and K. P. Basvarajappa in their articles present “Technological Innovation and Economic Development: Choices and Challenges for India”

There is growing awareness that technology-based innovations are major drivers of economic growth. If this were so, allocations for R and D should in some form be linked to major areas of economic activity [Chandrashekar 1995:2739-48]. Earlier studies had looked at R and D allocations to various areas and sectors and linked these allocations to the sizes of the markets served by them. Based on these linkages some suggestions were made for technology initiatives [TIFAC/ IIMB 1995]. There have been significant changes in the industrial and

business environment of the country since then. In this paper we take a re-look at the trends in R and D expenditure in different areas between 1994-95 and 1999-2000 and link these expenditures with the economic importance of the areas. Based on this assessment some suggestions are made for fostering and supporting technological innovation that can lead to accelerated economic growth. (1) Earlier studies had taken into account only the R and D spending of a few clearly identified scientific and technical organisations in the government sector. The current study adopts a more comprehensive approach and includes money spent by all central government entities that relate to technology and R and D [Government of India 1994-95, 1999- 2000]. (2) Since information technology (IT) has become a major focus of attention it has been identified as a separate area and

As per my revive this articles examines the trends in R and D expenditure in different areas between 1994-95 and 1999-2000 linking this expenditure with the economic importance of the areas. Based on some suggestions are made for supporting technological innovation that can lead to accelerated economic growth. Filled the gaps between needs and technology and creating the right conditions for large-scale diffusion of such technologies is one of the great challenges facing Indian.

2.3.9 Jon Pietruszkiewicz in their articles present “What are the Appropriate Roles for Government in Technology Deployment?”

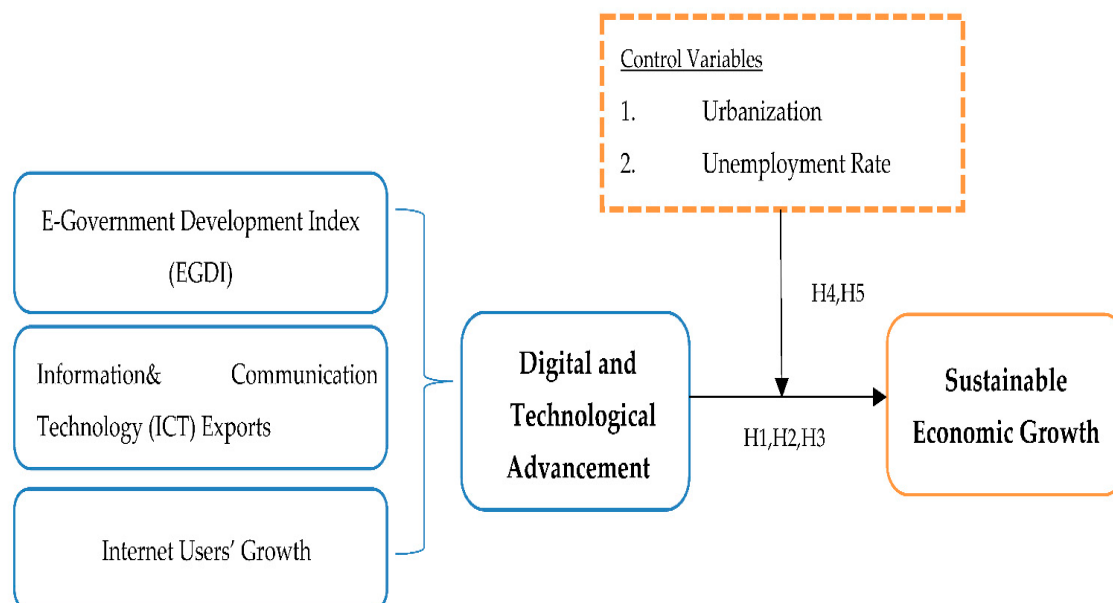
Technology leadership is broadly believed to be a key driver in maintaining our strong economy and global economic leadership. Technology is also an essential ingredient in our country’s ability to achieve important public objectives and government missions in areas such as public health, environmental protection, defences, space, and energy. The federal government has a long history of involvement in the successful development and commercialization of technology. Many of the country’s current technology leadership positions (e.g., aerospace, computers, the Internet, or medicine) came directly from government programs. In recent times, however, questions have arisen regarding the appropriate role for government in technology development and deployment. The drive to balance federal budgets, in combination with a strong economy, have led to calls for less government participation in scientific research, technology development, and especially technology deployment. While there is some degree of consensus that the federal government should continue to play a significant role in basic research and other long-term, high-risk research and development,^{1, 2, 3} less agreement appears to exist with respect to the federal government’s role in near-term research and deployment. Any government role appears to be susceptible labelled as “corporate welfare.” Industry knows it has the right technology when that technology, its performance and its cost, as well as other defining characteristics, meet a well-defined market need. * This may be a completely new need or just the need to improve upon existing alternatives. When the technology characteristics are not fully aligned with the perceived market requirements, or the market requirements are not easily discernible, industry is not inclined to invest in the technology development or commercialization.

As per my revive government can facilitate technology deployment to efficiently and effectively for public and for economy. Development and deployment of technology are often essential ingredients for accomplishment of such public benefit objectives. Technology

innovation and the welfare of the country are undeniably tightly linked. technology road mapping has become one of the more important and more widely used processes to formulate and communicate an informed vision of technology futures

2.3.10 Sainan Zhao ¹, Yichao Zhang ², Huma Iftikhar ³, Atta Ullah ^{3,4}, Jie Mao ⁵ and Tiantian Wang in their research present “Dynamic Influence of Digital and Technological Advancement on Sustainable Economic Growth in Belt and Road Initiative (BRI) Countries”

The digital and technological advancement initiated after the invention of the first electronic computer during the Second World War. The capacity for data storage and speed of data processing improved multiple times in the next decade. Numerous definitions of digital and technological advancement have been recommended over time. Digitalization was defined by as the influence of digital media on modern social life and digital communication. Digital and technological advancement integrates digital technologies in industry and produces financial development. It can be defined from an alternative viewpoint to grow business in a digital environment Development is determined by digital and technological progress only, so the role of digital and technological. advancement is very crucial for economic growth. This study is based on Solow’s neoclassical model as it measured the effect of digital and technological advancement on sustainable economic growth for the Asian countries of the BRI region. The structure of the study is represented in Figure



As per my review Asian countries should learn from China and European countries of BRI region regarding successful implementation of e-governance development systems. Developments in the e-government system will lead the country towards durable economic growth. To extract maximum positive consequences of trade on sustainable economic growth. e-commerce, and e-procurement should be promoted in Asian countries.

2.3.11 Remeikiene, R., Gaspareniene L., Fedajev, A., & Vebraite, V. in their articles present “The role of ICT development in boosting economic growth in transition economies.”

Having irreversibly transformed individuals’ habits, lifestyle, forms of work, communication, business and even governance processes, the ICT sector plays an important role in creating new jobs, new sources of income, new business models and reducing the cost of access to public services. There is a growing body of research in scientific literature focused on the analysis of the relationship between the ICT sector, ICT infrastructure development and investment in ICTs and national or regional economic growth. Based on the results of previous empirical studies, many researchers see ICT development as one of the key drivers of economic growth. Based on the report of the United Nations (2011), a positive macroeconomic impact of ICTs can arise as a result of:

- • higher productivity of the ICT sector itself.
-
- • ICT investment across the economy that raises labour productivity.
-
- • contribution of ICTs to business innovations and overall efficiency.

Research at the micro level helps to better understand what incentives drive investment in ICTs in businesses, how ICTs are incorporated into business processes and models, and how ICT development affects business performance, productivity, return on investment and flexibility of other factors (e.g., organizational structure, human resources, customer interface, etc.) The studies focused on the relationship between ICT development and economic growth in developed economies largely confirm the existence of a positive relationship between these variables.

As per my review To contribute to the investigation of this important issue, this paper is dealing with assessment of ICT development impact on economic growth. The impact of Internet users number is the largest and statistically significant, indicating that Internet penetration should be one of the top priority in the development strategies in transition countries. Especially Internet, can boost economic growth through fostering the technology diffusion and innovation, usage of e-government and e-commerce, improving decision-making in firms, households and economy, increasing demand, lowering the production costs, transforming the structure of economy and foreign trade.

2.3.12 Anaeto, F. C.*, Asiabaka, C. C., Ani, A. O., Nnadi, F. N., Ugwoke F. O., Asiabaka I. P., Anaeto, C. A. and Ihekeronye N. in their journal present “The roles of science and technology in national development”

Science and technology have been central in the progress and development of virtually all the nations of the world. It has contributed immensely in all sectors of the economy. Science and technology are intimately connected with development because; they have historical record of bringing advances that have led to healthier, longer, wealthier and more productive lives and they are key ingredients to solutions to the most serious poverty alleviation and economic development challenges that we currently face and are likely to face in the future. The many ways in which science and technology impact poverty alleviation across various sectors and economic growth merit attention. Science and technology have been central in the progress made to date in the fight against poverty and in stimulating economic growth. Advances in scientific knowledge and its application have helped slow the trend of high fertility, high mortality and led to increasingly better health for many people world over. Advance in science and technology have facilitated higher yields, greater efficiency and greater nutritional content in the world's food supply.

As per my revive Science and technology are tragically important to economic opportunity and growth. it became important to stimulate demand for technology from private sector We must ensure adequate development of manpower in science and technology to guarantee the efficient utilization of abundant natural resources and reduce the drain on our treasury and independence on outside sources for industrialization.

2.3.13 Jeffrey D. Sachs and John W. McArthur in their research present “Technological Advancement and Long-Term Economic Growth in Asia”

We are living in an age of remarkable technological change that is forcing us to think very hard about the linkages between technology and economic development. The harder we think about it, the more we realize that technological innovation is almost certainly the key driver of long-term economic growth. Economic theory offers a series of textbook approaches to understanding economic change. One of the first was initiated.

in 1776 by Adam Smith (Smith 1981), who emphasized the role of the division of labour in promoting rising output per person. He stressed that increasing specialization, mediated mainly by market forces, would lead to rising efficiency in production, and therefore to rising living standards. Smith focused on the role of market institutions efficiency in transactions, and effective property rights in promoting high levels 158 Jeffrey D. Sachs and John W. McArthur of economic well-being. Theoretical models stress that there are two basic modes of advancing technology. Economic theory demonstrates that if one economy is a technological innovator while another economy is a technology adopter, the innovator will maintain a lead in income per capita relative to the adopter. The income gap between the two economies persists over time even though the technology adopter ends up incorporating

all the technological advances made by the innovator. It does so, but only with a lag, and the persisting lag in technology translates into a persisting gap in income levels in favour of the innovator.

As per my revive There is no doubt that Asia has everything that it needs to become a central site of science-based innovation in the twenty-first century world economy. As the region progresses, we predict that one of twenty-first-century's biggest transitions will occur when both China and India begin to make dramatic contributions to global science and technology. innovation once economies reach a high level of development.

2.3.14 Andrew Reamer in their article present “The Impacts of Technological Invention on Economic Growth”

economists and economic historians have sought to improve their understanding of the role of technological invention in economic growth. As in many fields of inventive endeavour, their efforts required time to develop and mature. While the GDP and the rest of the national income accounts may seem to be arcane concepts, they are truly among the great inventions of the twentieth century.

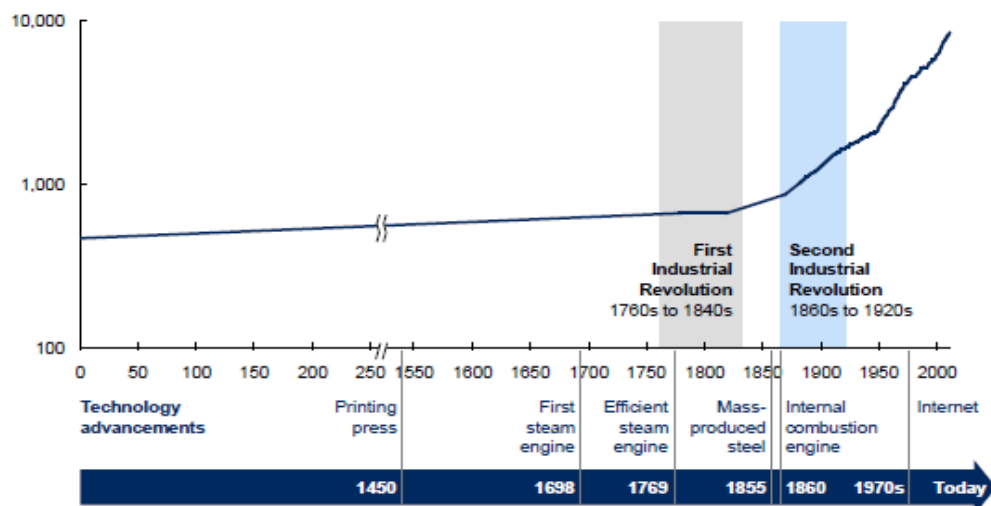
Much like a satellite in space can survey the weather across an entire continent so can the GDP give an overall picture of the state of the economy. It enables the President, Congress, and the Federal Reserve to judge whether the economy is contracting or expanding, whether the economy needs a boost or should be reined in a bit, and whether a severe recession or inflation threatens.

traditional GDP is more a measure of production, as currently valued by the market, than it is a more general measure of economic well-being. In recent years, the shortcomings of GDP has been an increasingly popular topic of policy and scholarly conversation. For the purposes of this paper, two types of issues are relevant.

Exhibit 1

Since the Industrial Revolution, the world has experienced an unprecedented rise in economic growth that has been fueled by innovation

Estimated global GDP per capita
\$



As per my revive. The industrial revolution increase economic development which is done with the help of technology Progress on some of the oldest and toughest challenges

associated with computers, robots, and other digital gear was gradual for a long time. Faster innovation-led growth is generally associated with higher turnover rates, i.e., higher rates of creation and destruction, of firms and jobs. Liberalized trade stimulates innovation by promoting competition and by increasing the market for successful innovations.

2.3.15 Sharmistha and Richard Grabowski in their research paper present “Economic Development and the Role of Agricultural Technology”

Initially development economics concentrated on questions concerning the industrialization process. The main questions had to do with how to modernize through overcoming the constraints of traditional society. many of these models it was presumed that the traditional sector was characterized by surplus labour. That is, there was so much labour in this sector that it could be withdrawn and put to productive work in the modern sector without any fall in output in the traditional sector (marginal product of labour in the traditional sector is zero). In effect, “free growth” was possible through mobilization of labour for modern production. After these developments, agriculture disappeared from general models aimed at analysing economic growth and development. Instead, much of the literature concerned with agriculture concentrated on analysing productivity growth in the traditional, agricultural sector. One can summarize much of this discussion in the following way. Agriculture’s role in the development process can be seen as a supplier of resources, most importantly labour, for the expansion of modern manufacturing. If disequilibrium exists such that the marginal product of labour in manufacturing exceeds that in agriculture (the marginal product of labour in agriculture need not be zero as Lewis originally argued), then transferring labour from the agricultural sector to the manufacturing sector will enhance overall growth. it is hypothesized that better agricultural technology which leads to improvements in agricultural productivity is a pre-condition for and has a significant impact on long-run growth, where growth is defined as the average rate of change of real per capita GDP. Biochemical technologies generally require an intensification of fertilizer usage while mechanical innovations require increased usage of implements such as tractors, often combined with increased fertilizer use.

As per my revive importance of a modernized agricultural sector for economic growth and development. The hypothesis was that improved technology or modernization of the agricultural sector improves wellbeing. Indeed, the measures of agricultural modernization seem to have statistically significant positive effects on human development. complement the recent theoretical work incorporating the agricultural sector into multi-sector growth models.

2.3.16 Ricardo Hausmann and José Domínguez in their research present “Knowledge, Technology and Complexity in Economic Growth”

The application of complexity science tools to the study of society allows for the analysis of phenomena that have been hard to identify and analyses with more traditional tools, especially in the field of Economics, which in the absence of these tools has tended to work with relatively low dimensional representations of reality. But the increasing availability of more detailed

information of social phenomena makes it particularly useful to use tools that can exploit this informational richness. This opens fascinating new horizons on almost all fields of knowledge in the social sciences. However, the mechanisms through which technology is developed, adopted and used in production are complex. Their more detailed analysis can allow for new findings that could have important impacts in many areas of policy, including science policy, research and development, industrial policy, and both national and regional development policies.



As per my review the development of science, technology, innovation and production require both codified and tacit knowledge but the codifiable component of science and technology get registered, technology and innovation (STI) would benefit enormously from a deeper and more detailed understanding of the 6-fold multiplex that this research wants to uncover.

2.4 research gap found in economic development due to technology.

The role of technology development in the Indian economy is a topic of great interest and ongoing research. Despite the numerous studies that have been conducted on the topic, there remains a gap in our understanding of the full impact of technology development on the Indian economy.

One area where more research is needed is the measurement of the actual impact of technology on the Indian economy. While there have been some studies that have attempted to measure the impact of technology, the results have been mixed and there is a need for more robust and reliable data to be collected and analysed.

Another area where research is needed is in understanding the barriers to technology adoption in India. Despite the potential benefits of technology, there are a few factors that are limiting its uptake in India, such as lack of infrastructure, lack of skilled personnel, and limited access to capital. Understanding these barriers and developing strategies to overcome them will be critical to fully realizing the potential benefits of technology for the Indian economy.

Additionally, more research is needed to understand the role of government in promoting technology development in India. While the government has taken a number of steps to encourage technology development, it is unclear how effective these efforts have been, and what more can be done to support the growth of the technology sector in India.

Finally, there is a need for more research on the impact of technology on different segments of the population in India. While technology has the potential to benefit all segments of society, it is important to understand how it is impacting different groups, such as rural communities, women, and marginalized populations, to ensure that the benefits are being distributed equitably.

In conclusion, there is a significant research gap in our understanding of the role of technology development in the Indian economy. Addressing these gaps will require ongoing research and collaboration between academics, policy makers, and industry experts. technology development in the Indian economy has been widely researched in recent years, but there is still a gap in the understanding of how technology can be leveraged to drive economic growth and improve the standard of living for the country's population. Some of the key areas where further research is needed include:

Impact of technology on employment: There is a need for further research to understand the effect of technology on employment in the Indian economy, particularly in the context of automation and the transition to a digital economy.

Technological innovation and entrepreneurship: Despite the presence of a large pool of talented entrepreneurs and engineers in India, the country has yet to fully leverage its potential for technological innovation and entrepreneurship. Further research is needed to understand the barriers to entrepreneurship and innovation, and how these can be overcome.

Bridging the digital divide: India is a country of vast economic and social disparities, and the digital divide remains a significant challenge. Further research is needed to understand how technology can be leveraged to bridge this divide and ensure that the benefits of technology are accessible to all sections of society.

Public-private partnerships: To truly leverage technology for economic growth, it is important to establish strong public-private partnerships that encourage investment in technology, research and development, and innovation. Further research is needed to understand how these partnerships can be structured and optimized to drive growth in the technology sector.

Regulation of technology: The rapid pace of technological change has raised important questions about the role of government in regulating and managing technology development in the country. Further research is needed to understand the impact of different regulatory frameworks on technology development, and how to ensure that regulations promote innovation while protecting public interests.

Technology-led entrepreneurship: India is a hub for technology startups, but there is a need for more research to understand the factors that contribute to their success and what can be done to support their growth.

Measuring the impact of technology: There is a need for more robust methods to measure the impact of technology on the Indian economy. This includes not just its contribution to GDP, but also its impact on different sectors and the well-being of individuals.

Sustainability of technology development: The rapid pace of technological change means that there is a need for research to understand how to ensure that the benefits of technology are sustained over the long-term, including its impact on the environment and natural resources.

REAFRANCE

1 MR ERIK J DE BRUIJN AND HARM-JAN BRUIJN, INTERNATIONAL JOURNAL OF INNOVATION AND TECHNOLOGY MANAGEMENT, VOLUME – 9, ISSUES -2012

2 I. A. MAGOMEDOV H A MURZAVE AND A M BAGOV, IOP CONFERENCE SERIES MATERIAL SCIENCE AND ENGINEERING, VOLUME – 862, ISSUE -2020

3 AHMED MUAYAD YOUNUS, ACADEMIC JOURNAL OF DIGITAL ECONOMICS AND STABILITY VOLUME 10, ISSUE -2021

4 HULYA KESICI CALISKAN, WORLD CONFERENCE ON TECHNOLOGY, INNOVATION AND ENTREPRENEURSHIP ISTANBUL UNIVERSITY, BAYEZID, FAITH, ISTANBUL 34452, TURKEY

5 DHANSHRI M. PATIL HON. SHRI. ANNASHEB DANGE ARTS, COMMERCE, SCIENCE MAHAVIDYALAYA, HATKANANGALE. GLOBAL MEDIA JOURNAL – INDIAN EDITION, VOLUME. -2, ISSUE NO-.2

6 IVO DE LOO AND LUC SOETE MAASTRICHT ECONOMIC RESEARCH INSTITUTE ON INNOVATION AND TECHNOLOGY, UNIVERSITY OF MAASTRICHT ISSUE -1999

7 DEBDAS BANERJEE ECONOMIC AND POLITICAL WEEKLY, VOLUME -33, ISSUE NOV-2020

8 S. CHANDRASHEKAR AND K. P. BASVARAJAPPA, ECONOMIC AND POLITICAL WEEKLY, VOLUME -36, ISSUE- 2001

9 JON PIETRUSZKIEWICZ, NATIONAL RENEWABLE ENERGY LABORATORY, NREL IS A U.S. DEPARTMENT OF ENERGY LABORATORY, MIDWEST RESEARCH INSTITUTE • BATTELLE • BECHTEL.

10 SAINAN ZHAO 1, YICHAO ZHANG 2, HUMA IFTIKHAR 3, ATTA ULLAH 3,4, JIE MAO 5 AND TIAN TIAN WANG, VOLUME -14, ISSUE -2022

11 REMEIKIENE, R., GASPARENIENE L., FEDAJEV, A., & VEBRAITE, V. JOURNAL OF INTERNATIONAL STUDIES VOLUME -14, ISSUE -2021

12 ANAETO, F. C. *, ASIABAKA, C. C., ANI, A. O., NNADI, F. N., UGWOKÉ F. O., ASIABAKA I. P., ANAETO, C. A. AND IHEKERONYE N. DEPARTMENT OF AGRICULTURAL EXTENSION, FEDERAL UNIVERSITY OF TECHNOLOGY, VOLUME -3, ISSUE 2016

13 JEFFREY D. SACHS AND JOHN W. MCARTHUR AT COLUMBIA UNIVERSITY ISSUE- 2002

14 ANDREW REAMER TECHNOLOGICAL CREATIVITY AND ECONOMIC PROGRESS, NEW YORK: OXFORD UNIVERSITY PRESS, ISSUE 2014

15 SHARMISTHA AND RICHARD GRABOWSKI, SOUTHERN ILLINOIS UNIVERSITY CARBONDALE, ISSUE - 2006